

VENKATRAMAN R

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Resume Headline

Data Scientist — Python, SQL, Machine Learning, Deep Learning, Generative AI — Actively Seeking Data Science / ML / AI Roles

Summary

Entry-level Data Scientist fluent across the ML lifecycle — statistics, algorithms, applied mathematics, modelling, explainability, analytics, and deployment — demonstrated through 8 production-grade projects shipped end to end. Strong in Python, SQL, machine learning (XGBoost, Random Forest, Scikit-Learn), deep learning (TensorFlow, Keras, LSTM), and generative AI (LangChain, FAISS, RAG); deployed models reach AUC-ROC 0.97 with audit-ready SHAP explainability. Actively seeking Data Scientist, Machine Learning Engineer, or AI Engineer roles.

Skills

Programming: Python, SQL

Math & Statistics: Statistics, Linear Algebra, Calculus, Probability, Algorithms, Hypothesis Testing

Machine Learning: Scikit-Learn, XGBoost, Random Forest, SMOTE, SHAP, Optuna, Cross-Validation, Feature Engineering

Deep Learning: TensorFlow, Keras, LSTM, Neural Networks, RNN, CNN

GenAI / LLMs: LangChain (LCEL), FAISS, RAG, OpenAI API, Groq Llama-3.3-70B, HuggingFace, Prompt Engineering, NLP

Analytics & Visualization: Pandas, NumPy, Power BI (DAX), Plotly, Matplotlib, Seaborn, Streamlit, SQLite, Parquet, Power Query

Tools: Git, GitHub, Jupyter, VS Code, Google Colab, Advanced Excel

Experience / Internship

Data Science Trainee

Jan 2026 – Jun 2026

Boston Institute of Analytics — Professional Certification Program

- Built FraudGuard, an XGBoost classifier with Optuna tuning (100 trials) and SMOTE balancing on imbalanced mobile-transaction data, reaching AUC-ROC 0.97, F1 0.92; encoded fraud-cost asymmetry via F2-score threshold optimization and added SHAP explanations for audit compliance
- Designed BankShield, an end-to-end Random Forest pipeline (EDA to feature engineering to SMOTE to threshold tuning to deployment) reaching AUC-ROC 0.97, precision 0.94, recall 0.89; identified transaction velocity as more predictive than amount and deployed a Streamlit app with single/batch CSV scoring

Projects

AskMyData — GenAI Dataset Q&A Application — Python, LangChain (LCEL), FAISS, Groq Llama-3.3-70B, OpenAI, Streamlit

- Built a LangChain LCEL agent with a natural-language interface that auto-selects Pandas or SQL to answer questions across 4 source formats (CSV, Excel, JSON, SQLite), using FAISS to retrieve per-column schema context for accuracy
- Engineered a secure execution sandbox with whitelisted builtins and auto-retry, multi-turn memory, auto-generated Plotly visualizations per query, and a live Streamlit Cloud deployment

SalesPulse — Power BI Sales Intelligence Dashboard — Power BI, DAX, Power Query, Star Schema

- Authored 44 custom DAX measures (YoY, MTD/QTD/YTD, customer LTV, profit margin) over a star schema on \$12.63M of Global Superstore data, with executive, regional, product, and customer-segment drill-throughs

SalesPulse Forecaster — LSTM 12-Month Sales Forecasting — TensorFlow, Keras, LSTM, SQLite, Streamlit

- Built a 2-layer deep learning LSTM network (128 to 64 units) generating recursive 12-month forecasts with confidence bands, backed by a SQLite pipeline and a Streamlit upload-to-export interface

DataHunt — Automated ETL Pipeline — Python, Pandas, Parquet, openpyxl

- Designed a modular pipeline (Extractor to Cleaner to Storage to Reporter) processing 51,000+ records and outputting Parquet and Excel simultaneously, with independently swappable modules

Vehicle Maintenance Dashboard — Fleet Analytics — Advanced Excel

- Built a fleet analytics dashboard over 50,000 vehicle maintenance records with KPI tracking and conditional-formatting-driven alerts

Education

Bachelor of Engineering (B.E.) — Automobile Engineering

2021 – 2024

Vels Institute of Science, Technology & Advanced Studies (VISTAS), CGPA: 7.25

Certifications

- Professional Certification in Data Science with Artificial Intelligence — Boston Institute of Analytics (Jun 2026)